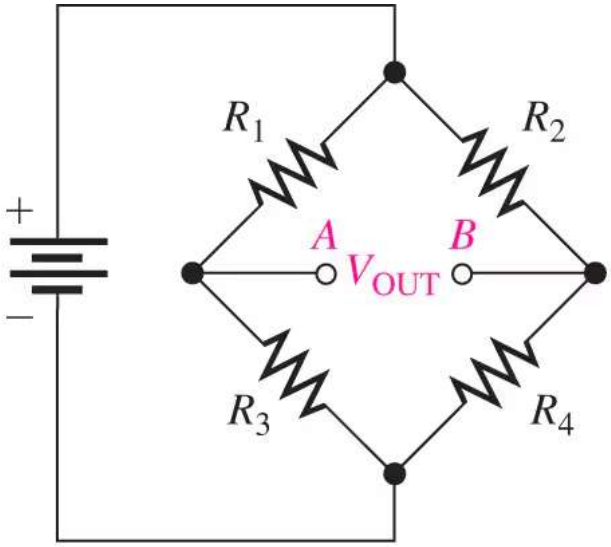
Sensing using Resistor Network-**Wheatstone's Bridge**

- Wheatstone's bridge is commonly used in applications that rely on **changes in resistance** to measure physical quantities like **force, pressure, or temperature.**



Balanced bridge condition:

- Output voltage between terminals A and B is **equal to zero** ($V_{out} = 0V$)
- The products of resistances in the opposite diagonal arms are equal. ($R_1 X R_4 = R_2 X R_3$) or ($R_1/R_3 = R_2/R_4$)

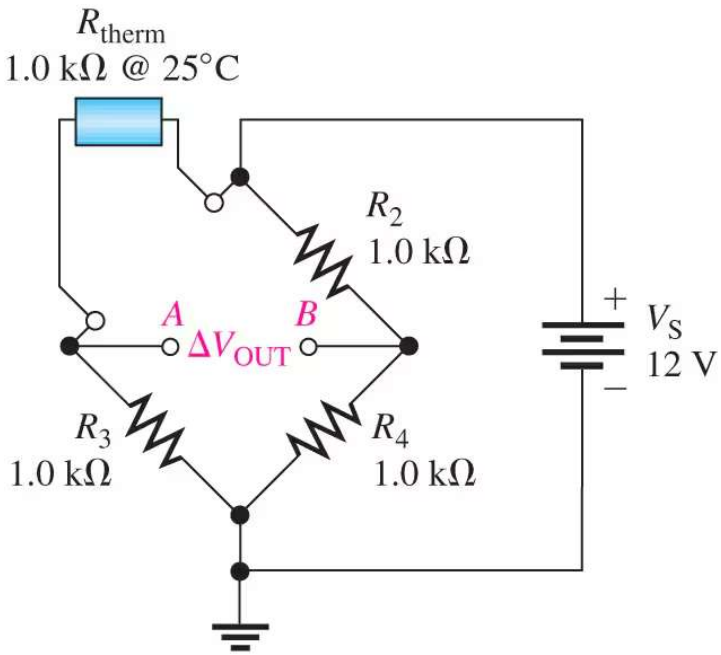
Unbalanced State:

A bridge circuit that is in the **unbalanced state** as indicated by a voltage across the bridge that is proportional to the amount of deviation from the balanced state.

$(V_{out} \neq 0V)$

An unknown resistance in one position can be accurately measured using the balanced state of the bridge.

Measuring Temperature using Wheatstone's Bridge



A Bridge Circuit for Measuring Temperature:

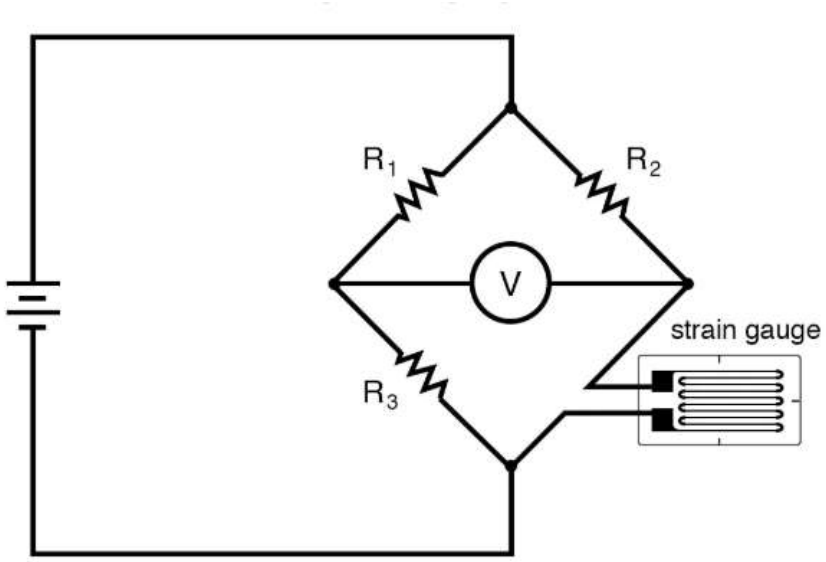
- It is balanced at a reference temperature
- It becomes unbalanced at a measured temperature.

- For example, let's say the bridge is to be balanced at 25° C . A thermistor will have a known value of resistance at 25° C .

At 50° C, resistance drops(say, 900 Ω) producing $V_{out} \neq 0V$

Measuring strain using Wheatstone's Bridge

- The resistance of the strain gauge changes proportionally to the changes in the strain that it is measuring.
- If the bridge is balanced at a known point, then the amount of **deviation from the balanced condition**, as indicated by **the output voltage**, indicates the amount of change in the parameter(**strain**) being measured.
- Therefore, the **amount of imbalance** can determine the **value of the strain being** measured.



A **strain gauge** is a sensor whose measured electrical resistance varies with changes in strain.